ABU ADNAN SHAD

Computer Science & Engineering | AI/ML Research | Software Development

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LinkedIn | GitHub | Codeforces

RESEARCH PROFILE

Computer Science and Engineering graduate pursuing research in **artificial intelligence, machine learning, and trustworthy real-world AI applications**, with current work on **imbalanced chronic kidney disease screening using engineered CatBoost models**. Research interests include **deep learning, computer vision, large language models, vision-language models, graph neural networks, and agentic AI**. Combines research experience with competitive programming, algorithmic problem solving, and hands-on Android/web/mobile application development.

TECHNICAL EXPERTISE

Programming Languages: Python, C, C++, Java, Dart

CS Foundations: Data Structures & Algorithms, OOP, DBMS, OS, Software Engineering, Computer Networks, Theory of Computation

Mathematical Foundations: Linear Algebra, Calculus, Probability, Statistics

Application Development: HTML, CSS, Flutter (Dart), Android (Java), Firebase

Tools & Platforms: Git, GitHub, Google Colab, VS Code, Android Studio,

Machine Learning: Logistic Regression, Random Forest, SVM, XGBoost, CatBoost

ML / Data Libraries: scikit-learn, pandas, NumPy, Matplotlib, Seaborn, SHAP

Research Methods: Imbalanced classification, model benchmarking, threshold optimization, explainability with SHAP

Current AI Learning: TensorFlow, PyTorch, DNNs, LSTM, GRU, VGG16/19, Inception, MobileNet, ViT, YOLOv5/v6, R-CNN, Mask R-CNN, Transformers, BERT, LLaMA, VLMs, GNNs, Agentic AI

EDUCATION

Bachelor of Science in Computer Science and Engineering **2022–2026**
Leading University, Sylhet, Bangladesh
CGPA: 3.62/4.00

RESEARCH EXPERIENCE

Engineered CatBoost for Imbalanced Chronic Kidney Disease Screening *Undergraduate Thesis*
Python, CatBoost, scikit-learn, XGBoost, SHAP

Research problem: Developed a machine-learning screening pipeline for chronic kidney disease using 1,659 imbalanced patient records, addressing the challenge of reliable detection under class imbalance.

Approach: Led a two-student thesis under faculty supervision; engineered and evaluated a CatBoost model and benchmarked it against Logistic Regression, Random Forest, XGBoost, and SVM.

Evaluation & Results: Applied threshold optimization and SHAP-based interpretation, achieving 0.9605 sensitivity, 0.7660 balanced accuracy, 0.8210 ROC-AUC, and 0.9767 PR-AUC on the held-out test set.

Research output: Manuscript in preparation for conference submission.

SELECTED TECHNICAL PROJECTS

ClubMate | **Android, Firebase, Cloudinary**

Problem: University clubs often manage messaging, announcements, membership, and private communication across disconnected channels.

Solution: Team-developed an end-to-end encrypted Android application that centralizes messaging, announcements, member management, role-based groups, notice boards, private channels, and encrypted media sharing.

Contribution: Contributed to research and development and performed application testing.

HostelMate | **Flutter, Dart, Firebase**

Problem: Manual management of hostel meals, shared expenses, shop dues, and monthly balances often causes calculation errors and disputes.

Solution: Independently developed a role-based Android application with Firebase, administrator controls, secure data access, and automated monthly calculations.

CGPA Calculator | **Flutter, Dart, Mobile Development**

Problem: Manual semester and cumulative GPA calculation can be time-consuming and error-prone for students.

Solution: Independently developed a mobile application that accepts course credits and grades to calculate semester GPA and cumulative CGPA for academic planning.

RobGlass | **Arduino, Ultrasonic Sensors, Buzzer**

Problem: Visually impaired users need low-cost assistance for detecting nearby obstacles.

Solution: Co-developed Arduino-based assistive glasses with configurable ultrasonic detection and buzzer alerts; contributed to R&D, programming, and circuit design.

COMPETITIVE PROGRAMMING & PROBLEM SOLVING

250+ algorithmic problems solved on Codeforces and CodeChef across data structures, DP, and graph algorithms.

50+ online and university-level onsite programming contests, including ICPC and individual rated rounds.

TEACHING, OUTREACH & LEADERSHIP

Student Assistant, Machine Learning Laboratory (Fall 2025): Selected by the course instructor to support junior students with laboratory exercises and machine-learning concepts.

Publications and Newsletter Coordinator, IEEE Computer Society, Leading University (2024–2025): Coordinated publication, newsletter, and event-documentation activities.

Founding Member and Program Manager, Igniting Smiles Foundation (2022–Present): Coordinates aid initiatives for low-income communities and disaster relief in Bangladesh.

STEM Content Creator: Created and presented short educational videos through **180 Seconds with Adnan**.

HONORS & PROFESSIONAL AFFILIATIONS

Most Promising Volunteer Award, IEEE Computer Society, Leading University: 2023 & 2024

Student Member, Institute of Electrical and Electronics Engineers (IEEE): 2024–Present

Member, Leading University ACM Community: 2024–2025

STANDARDIZED TEST

IELTS Academic (September 19, 2026): Overall 6.5 | Listening 6.5 | Reading 6.0 | Writing 6.0 | Speaking 6.5